

Transcriptome-wide Association Studies

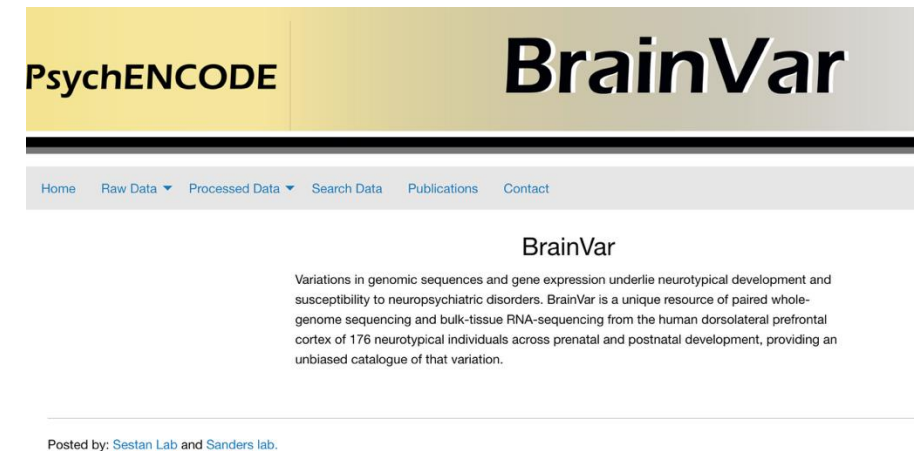
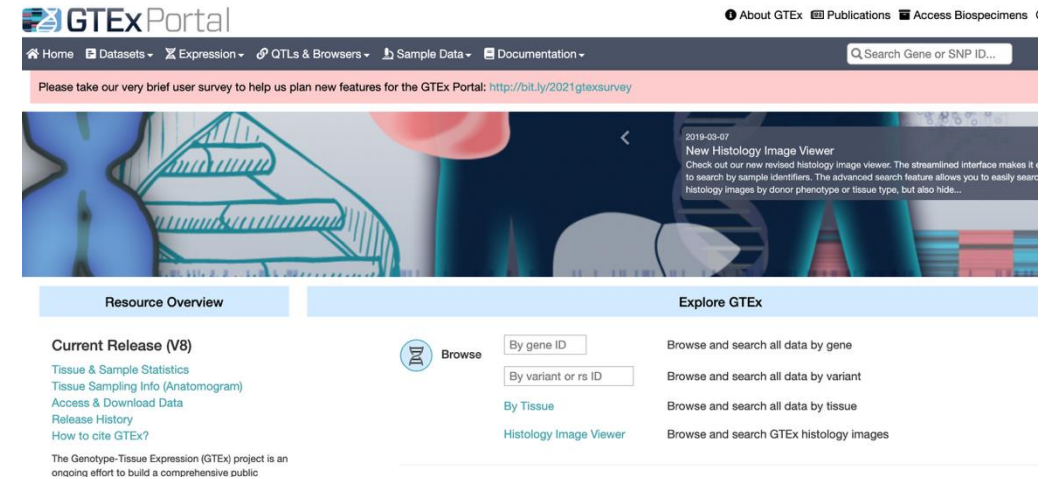
Lecture 2

Outline

- OTTERS: Leveraging eQTL Summary Data for TWAS
- Fine-mapping TWAS Results

TWAS Requires Reference Transcriptomic Data

- Leverage reference datasets possessing both genetic and transcriptomic data
- Transcriptome-wide Association Study (TWAS):
 - Exploit SNP-expression relationships in reference data to impute gene expression in GWAS study
 - Test association between imputed gene expression (GReX) and phenotype



Summary-level eQTL data are often publicly available



eQTL Catalogue

Expression and splicing QTLs recomputed from public datasets

[Project](#) | [Methods](#) | [Data access](#) | **[Studies](#)** | [License](#) | [Team](#) | [Release notes](#) | [Federated analysis](#)

List of studies included in the eQTL Catalogue

Bulk RNA-seq studies

Study ID	Publication	Cell types or tissues	Conditions	Samples	Donors	Raw expression data	Raw genotype data	QC report
QTS000001	Alasoo_2018	macrophages	IFNg, Salmonella, IFNg + Salmonella	336	84	EGAD00001003204 ; PRJEB18997	EGAD000010001147 ; PRJEB11752	link
QTS000002	BLUEPRINT	monocytes, neutrophils, CD4+ T cells		554	197	EGAD00001002671 ; EGAD00001002674 ; EGAD00001002675	EGAD00001002663	1,2
QTS000012	GENCORD	LCLs ¹ , fibroblasts, T cells		560	195	EGAD00001000425	EGAD00001000428	link
QTS000013	GEUVADIS	LCLs ¹		445	445	E-GEUV-1	1000 Genomes	link
QTS000016	HipSci	iPSCs ²		322	322	EGAD00001003529 ; PRJEB7388	EGAD000010001147 ; PRJEB11752	link

Expression Quantitative Trait Loci (eQTL) Analysis

- Consider the profiled gene expression levels as the quantitative trait E_g in the following single variant linear regression model:

$$E_g = \beta_0 + \alpha C + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

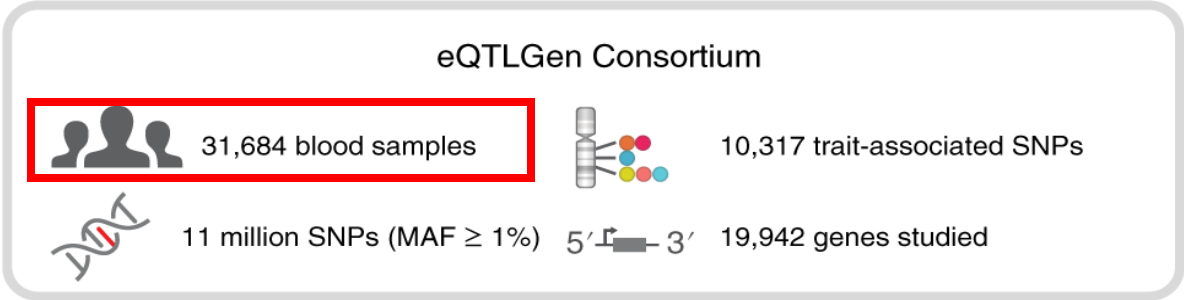
- X represents the genotype data (0, 1, 2) or dosage [0, 2] of the test SNP
 - C represents the confounding covariates or other environmental variables
 - ϵ represents the error term, other unknown factors
-
- eQTL: SNPs **significantly associated** with a gene expression quantitative trait ($H_0: \beta_1 = 0$ is significantly rejected) are referred as **expression Quantitative Trait Loci (eQTL)**

Key Components of Summary-level eQTL Data

Each row in a summary-level eQTL dataset usually represents a single association between a genetic variant (SNP) and gene expression, and includes:

- **SNP ID** (e.g., rsID)
- **Gene ID or symbol** (e.g., ENSG or HGNC symbol)
- **Chromosomal position** of the SNP and/or gene
- **Effect allele / non-effect allele**
- **Effect size** (e.g., beta)
- **Standard error**
- **P-value**
- **Minor allele frequency (MAF)** (optional)
- **Sample size**

Emerging large-scale summary-level expression reference



	Individuals
Genes	9 2 4 ... 7 6 3
	3 1 2 ... 1 2 4
	0 0 1 ... 2 0 0
	...
	3 2 7 ... 1 2 3

	Individuals
SNP counts	0 1 0 ... 1 0 0
	0 0 0 ... 0 0 1
	1 2 0 ... 0 1 0
	...
	0 0 1 ... 0 2 0

Individual-level data: not publicly available

eQTL analysis: **marginal** association between gene expression and each SNP separately

$$\begin{aligned}
 GE_g &= \beta_1 x_1 + \varepsilon_1 \\
 &\dots \\
 GE_g &= \beta_k x_k + \varepsilon_k
 \end{aligned}$$

Results: eQTL summary-level statistics

SNP	Gene	$\tilde{\beta}$
SNP 1	Gene 1	0.9
...	Gene 1	...
SNP k	Gene 1	0.2

Summary-level statistics (n = 31K): publicly available

Can we perform TWAS using eQTL summary-level statistics?

- allows TWAS to be applied in broader analysis settings
- increase TWAS power by using a reference data with much larger sample size
(summary-level eQTLGen $n = 31K$ vs.
Individual-level GTEx $n < 1000$)

Idea from Polygenic risk score (PRS) methods

- A PRS is a numerical estimate of an individual's genetic risk of developing [a certain trait or condition](#).
- It is calculated as the weighted sum of an individual's SNP genotypes ($X_1, \dots, X_k$):

$$PRS = \widehat{\beta}_1 X_1 + \dots + \widehat{\beta}_k X_k + \epsilon$$

- The GReX calculated in TWAS Step 2 is also the weighted sum of an individual's genotypes.

$$\widehat{GReX}_g = \widehat{\beta}_1 X_1 + \dots + \widehat{\beta}_k X_k + \epsilon$$

- $\widehat{GReX}_g$ can be viewed as a PRS [with gene expression as the trait](#).

Idea from Polygenic risk score (PRS) methods

GWAS summary-level statistics

SNP	Trait	$\tilde{\beta}$
SNP1	Height	0.9
...	Height	...
SNP k	Height	0.2

$$Y = \beta_k X_k + \varepsilon_k$$



PRS methods

$$PRS = \hat{\beta}_1 X_1 + \dots + \hat{\beta}_k X_k + \epsilon$$

eQTL summary-level statistics

SNP	Gene	$\tilde{\beta}$
SNP 1	Gene 1	0.9
...	Gene 1	...
SNP k	Gene 1	0.2

$$GE_g = \beta_k X_k + \varepsilon_k$$



PRS methods

$$\widehat{GReX}_g = \hat{\beta}_1 X_1 + \dots + \hat{\beta}_k X_k + \epsilon$$

Stage I: Training Imputation Models (Estimate $\mathbf{w}$) from Summary-level eQTL Data

- **Goal:** Training gene expression prediction model with summary-level eQTL data:

$$E_g = X\mathbf{w} + \varepsilon$$

- **Summary-level eQTL Data**

- Effect sizes and p-values from single-variant regression models used in eQTL analysis:

$$E_g = X_j \widetilde{\mathbf{w}}_j + \epsilon, \quad \epsilon \sim N(0, \sigma_\epsilon^2 \mathbf{I}), \quad j = 1, \dots, m.$$

$X_j, \widetilde{\mathbf{w}}_j$: Genotype vector, marginal effect size for cis-eQTL j

- Reference LD panel of cis-eQTLs from population of similar ancestry such as the 1000 Genomes Project or GTEx WGS data

Training Imputation Models (Estimate w)

- Considered four complementary summary-level training methods used in traditional **Polygenic Risk Score (PRS)** analysis of complex traits
 - P-value thresholding ($P+T$)
 - Summary-statistic LASSO (*lassosum*)
 - Summary-level Bayesian Dirichlet process regression (*SDPR*)
 - PRS with continuous shrinkage prior (*PRS-CS*)

P-value thresholding ($P+T$)

- Given eQTL summary data from:

$$E_g = X_j \widetilde{w}_j + \epsilon, \epsilon \sim N(0, \sigma_\epsilon^2 I), j = 1, \dots, m.$$

- Pick a p-value threshold (e.g., 0.001) to filter out eQTL with p-value less significant than the threshold
- Pick a R^2 threshold (e.g., 0.3) to select the top significant eQTL among all eQTL with genotype R^2 greater than the threshold
- Take the marginal eQTL effect sizes $\widetilde{w}_j$ as variant weights for Stage-II gene-based association test

PLINK Tool: <https://zzz.bwh.harvard.edu/plink/>

PRS Methods using Summary Data

- Recall multiple linear regression model for gene expression prediction (or deriving polygenic risk score):

$$E_g = X\mathbf{w} + \varepsilon$$

- Assume E_g and columns of X are centered with mean 0 and standardized with variance 1

- GWAS/eQTL summary data from marginal single variant linear regression model:

$$E_g = X_j \widetilde{\mathbf{w}}_j + \epsilon, \epsilon \sim N(0, \sigma_\epsilon^2 \mathbf{I}), j = 1, \dots, m$$

- Goal:** Estimates $\widehat{\mathbf{w}}$ in the multiple linear regression model with marginal effect size

estimates $\left\{ \widetilde{\mathbf{w}}_j, = \frac{E_g^T X_j}{n}, j = 1, \dots, m \right\}$, marginal Z-score statistics $\{Z_j, j = 1, \dots, m\}$,

sample size n , reference genotype correlation matrix (LD) R

- Note:** $\widetilde{\mathbf{w}} = \frac{E_g^T X}{n}$; $R = \frac{X^T X}{n}$; $Z = \widetilde{\mathbf{w}} \sqrt{n}$

Summary-statistic LASSO (*lassosum*)

- Multiple linear regression model for gene expression prediction:

$$E_g = X\mathbf{w} + \varepsilon$$

- LASSO estimates $\mathbf{w}$ in the multiple linear regression model by minimizing the following loss function with a penalty parameter λ :

$$f(\mathbf{w}) = \frac{\mathbf{E}_g^T \mathbf{E}_g}{n} + \mathbf{w}^T \mathbf{R} \mathbf{w} - 2\mathbf{w}^T \tilde{\mathbf{w}} + 2\lambda \|\mathbf{w}\|_1$$

- LassoSum takes $R = (1 - s)R_r + sI, 0 < s < 1; R_r = \frac{X_r^T X_r}{n_r};$

Mak et al., Genetic Epidemiology 2017

<https://choishingwan.github.io/PRS-Tutorial/lassosum/>

Summary-level Bayesian Dirichlet Process Regression (*SDPR*)

- Multiple linear regression model for gene expression imputation

$$\mathbf{E}_g = \mathbf{X}\mathbf{w} + \varepsilon$$

- Note that the least square estimate based on the multiple linear regression model is

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{E}_g^T \mathbf{X} = \left(\frac{\mathbf{R}^{-1}}{n} \right) \tilde{\mathbf{w}} \mathbf{n} = \mathbf{R}^{-1} \tilde{\mathbf{w}}; \quad \tilde{\mathbf{w}} = \mathbf{R} \hat{\mathbf{w}}$$

- Nonparametric Dirichlet Process Prior is assumed for

$$\mathbf{w}_j \sim \mathcal{N}(\mathbf{0}, \sigma_w^2), \sigma_w^2 \sim \mathcal{DP}(\mathbf{H}, \alpha),$$

- Base distribution $\mathbf{H} = \mathcal{IG}(\mathbf{a}_0, \mathbf{b}_0)$
- Concentration parameter α controlling the shrinkage of the distribution on σ_w^2 toward $\mathbf{H}$.

Summary-level Bayesian Dirichlet Process Regression (*SDPR*)

- SDPR uses marginal eQTL effect sizes $\tilde{w}$ as input data

$$\frac{\tilde{w}}{c} | w \sim N \left(R w, \frac{R + n a I}{n} \right)$$

- Scale marginal effect size by a constant c to correct for deflation of summary statistics if double genomic control was applied for generating the summary data by meta-analysis, which can be estimated by SumHer (Speed D. et al. Nat. Genetics 2019)
- Assume correlation between two SNPs is $\frac{R_{ij}}{1+na}$ rather than R_{ij} , with parameter a to shrink the correlation between two SNPs;
 - I is an identify matrix
- Estimate w by MCMC

Zhou et al., PLoS Genet. 2021

<https://github.com/eldronzhou/SDPR>

PRS with continuous shrinkage prior (*PRS-CS*)

- PRS-CS assumes a normal prior for $\mathbf{w}_j$ and non-informative scale-invariant Jeffreys prior on residual variance σ_ϵ^2 in the multiple linear regression model

$$\mathbf{E}_g = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}), \quad p(\sigma_\epsilon^2) \propto \sigma_\epsilon^{-2}$$

- Priors:

$$w_j \sim N\left(0, \frac{\sigma_\epsilon^2}{n} \phi \psi_j\right); \psi_j \sim \text{Gamma}(a, \delta_j), \delta_j \sim \text{Gamma}(b, 1)$$

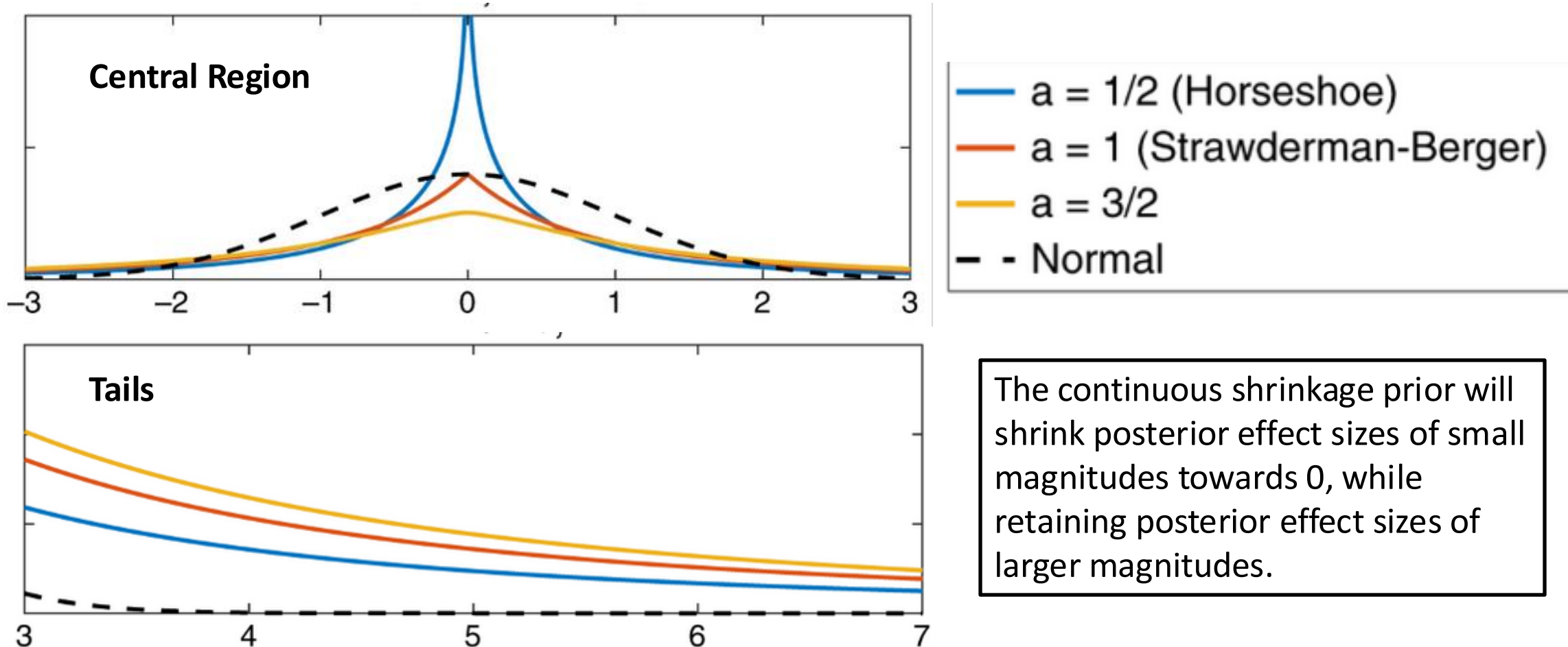
- ψ_j : local shrinkage parameter with a gamma-gamma prior distribution, Gamma(shape parameter, scale parameter)
- ϕ : global shrinkage parameter controlling the overall sparsity of $\mathbf{w}$

Ge et al. Nat. Commun. 2019

<https://github.com/getian107/PRScs>

PRS with continuous shrinkage prior (*PRS-CS*)

Marginal prior density on the effect size w_j with $\psi_j = 1$, $b = 0.5$, $a = (0.5, 1, 3/2)$



PRS with continuous shrinkage prior (*PRS-CS*)

- Posterior estimates of $\mathbf{w}$ can be obtained by using only summary-level eQTL data $\tilde{\mathbf{w}}$ and reference genotype correlation matrix $\mathbf{R}$ by Gibbs Sampler

$$\mathbf{w} \mid \sigma_{\epsilon}^2, \mathbf{\Psi}, \tilde{\mathbf{w}}, \mathbf{R} \sim \text{MVN} \left(\frac{n}{\sigma_{\epsilon}^2} \mathbf{\Sigma} \tilde{\mathbf{w}}, \mathbf{\Sigma} \right)$$

$$\mathbf{\Sigma} = \frac{\sigma_{\epsilon}^2}{n} (\mathbf{R} + \mathbf{\Psi}^{-1})^{-1}, \mathbf{\Psi} = \text{diag}(\psi_1, \dots, \psi_m)$$

TWAS Stage II: GReX Prediction and Testing

- Based on $\hat{\mathbf{w}}$, impute gene expression in GWAS cohort as $\widehat{GReX} = \mathbf{X}_{test}\hat{\mathbf{w}}$
 - $\mathbf{X}_{test}$: Genotype matrix in the test (GWAS) cohort
- Test association between $\widehat{GReX}$ and phenotype $\mathbf{Y}$, adjusting for covariates $\mathbf{Z}$, using regression model:

$$E[g(\mathbf{Y}|\mathbf{X}_{test})] = \boldsymbol{\beta}\widehat{GReX} + \boldsymbol{\alpha}'\mathbf{Z} = \boldsymbol{\beta}(\mathbf{X}_{test}\hat{\mathbf{w}}) + \boldsymbol{\alpha}'\mathbf{Z}$$

- **TWAS tests** $H_0: \boldsymbol{\beta} = \mathbf{0}$ using score/Wald/LR test from regression model

TWAS Stage II: Testing with Summary-level GWAS Data

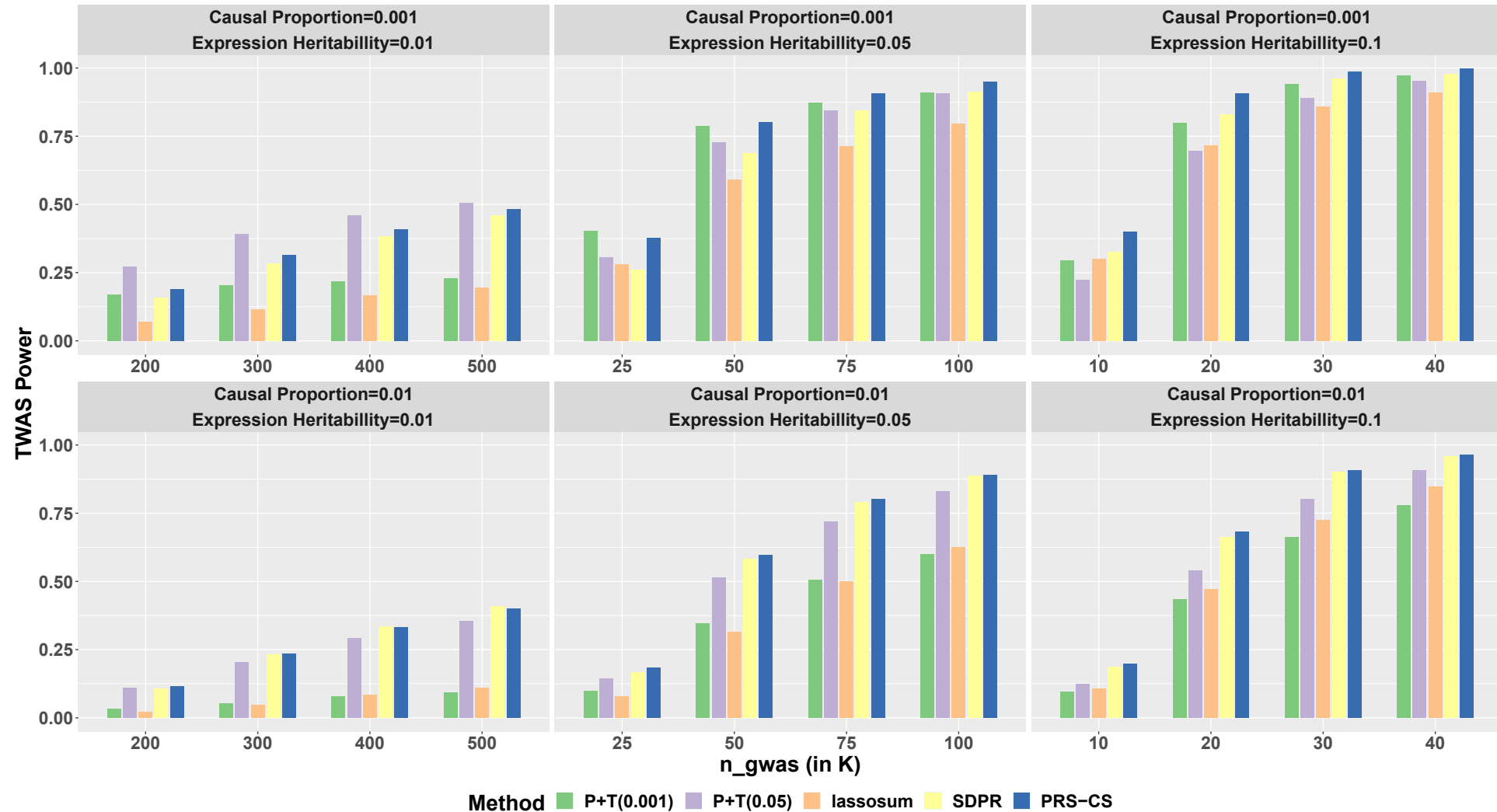
- With summary-level GWAS data (Z scores), one can instead test

$$Z_{g,FUSION} = \frac{\sum_{j=1}^J (\hat{w}_j Z_j)}{\sqrt{\hat{\mathbf{w}}' \mathbf{V} \hat{\mathbf{w}}}}$$

- Z_j : Single-variant Z score in GWAS for test SNP (“eQTL”) j .
- $\mathbf{V}$: Genotype correlation matrix, which can be approximated from reference genotype panel (e.g. 1000 Genome).
- Centered and standardized gene expression and genotypes were assumed for deriving estimates of eQTL weights.

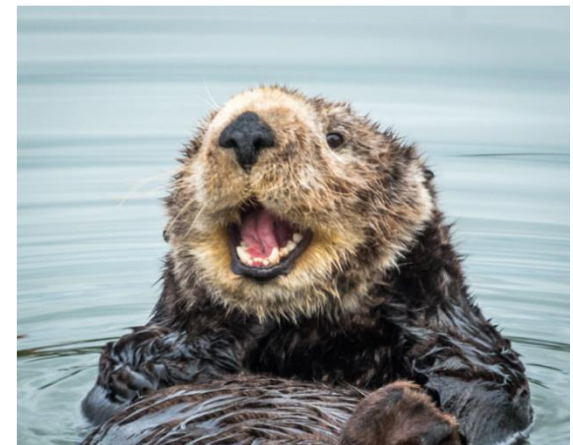
Which PRS method to use?

- **Classical:** **P+T**
(0.001) **P+T** (0.05)
- **Penalized Regression:**
lassosum
- **Bayesian Regression:**
SDPR, **PRS-CS**



OTTERS Motivation

- **Issue:** Performance using individual training method depends on the underlying genetic architecture of gene expression
 - Idea: Create an omnibus test by combining individual TWAS p-values together using **Aggregated Cauchy Association Test (ACAT)**
- **Omnibus Transcriptome Test** using **Expression Reference Summary** data (**OTTERS**)



OTTERS p-value by ACAT

- p_k : TWAS p-value with eQTL weights trained under k^{th} method
 - $\tan\{(0.5 - p_k)\}$ follows Cauchy under null
- $T_{OTTERS} = \frac{1}{K} \sum_{k=1}^K \tan\{(0.5 - p_k)\pi\}$, $\pi \approx 3.14$
 - Can be approximated by Cauchy distribution
- **OTTERS p-value:** $p_{OTTERS} \approx \frac{1}{2} - \frac{\{\arctan(T_{OTTERS})\}}{\pi}$

ACAT can aggregate correlated p-values to increase power and control Type I error for the same hypothesis test

- Examples:

- Example 1: ACAT improves Power

10^{-9}

10^{-3}

10^{-3}

10^{-3}

10^{-3}

ACAT: $5 \times 10^{-9} < 2.5 \times 10^{-6}$
genome-wide significant

- Example 2: ACAT controls Type-I error

10^{-6}

10^{-3}

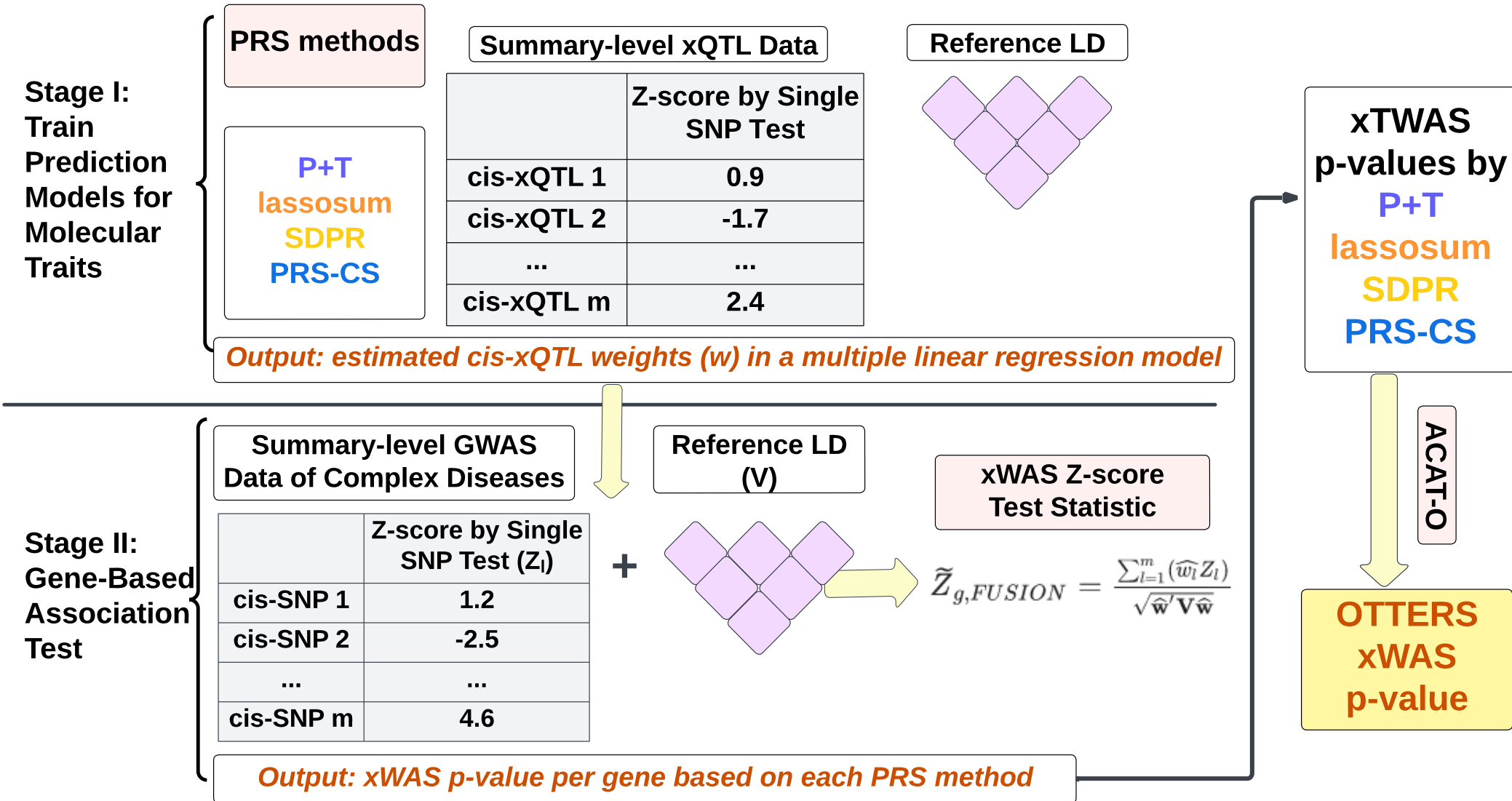
10^{-3}

10^{-3}

10^{-3}

ACAT: $5 \times 10^{-6} > 2.5 \times 10^{-6}$
Not significant

OTTERS framework

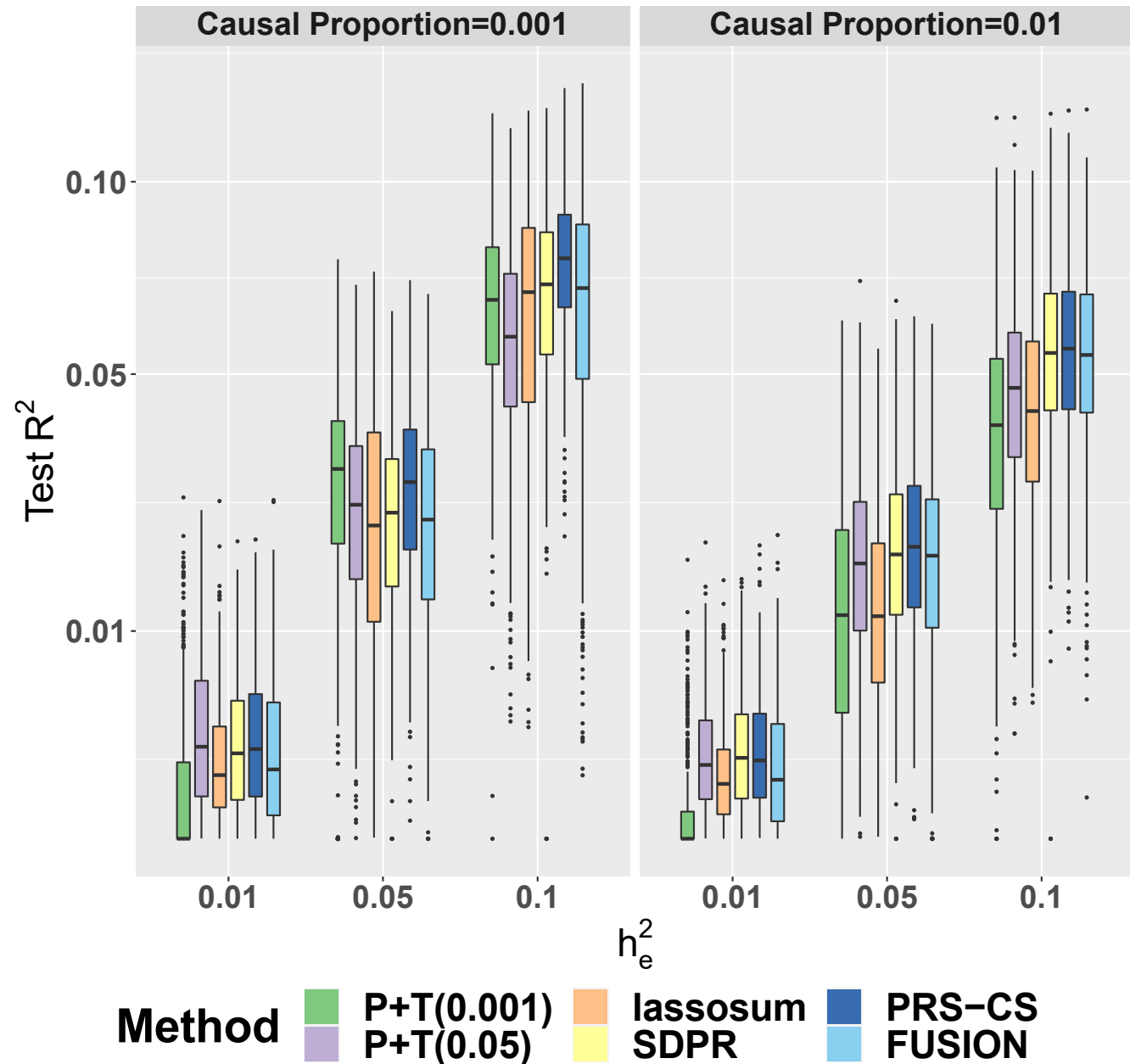


Simulations

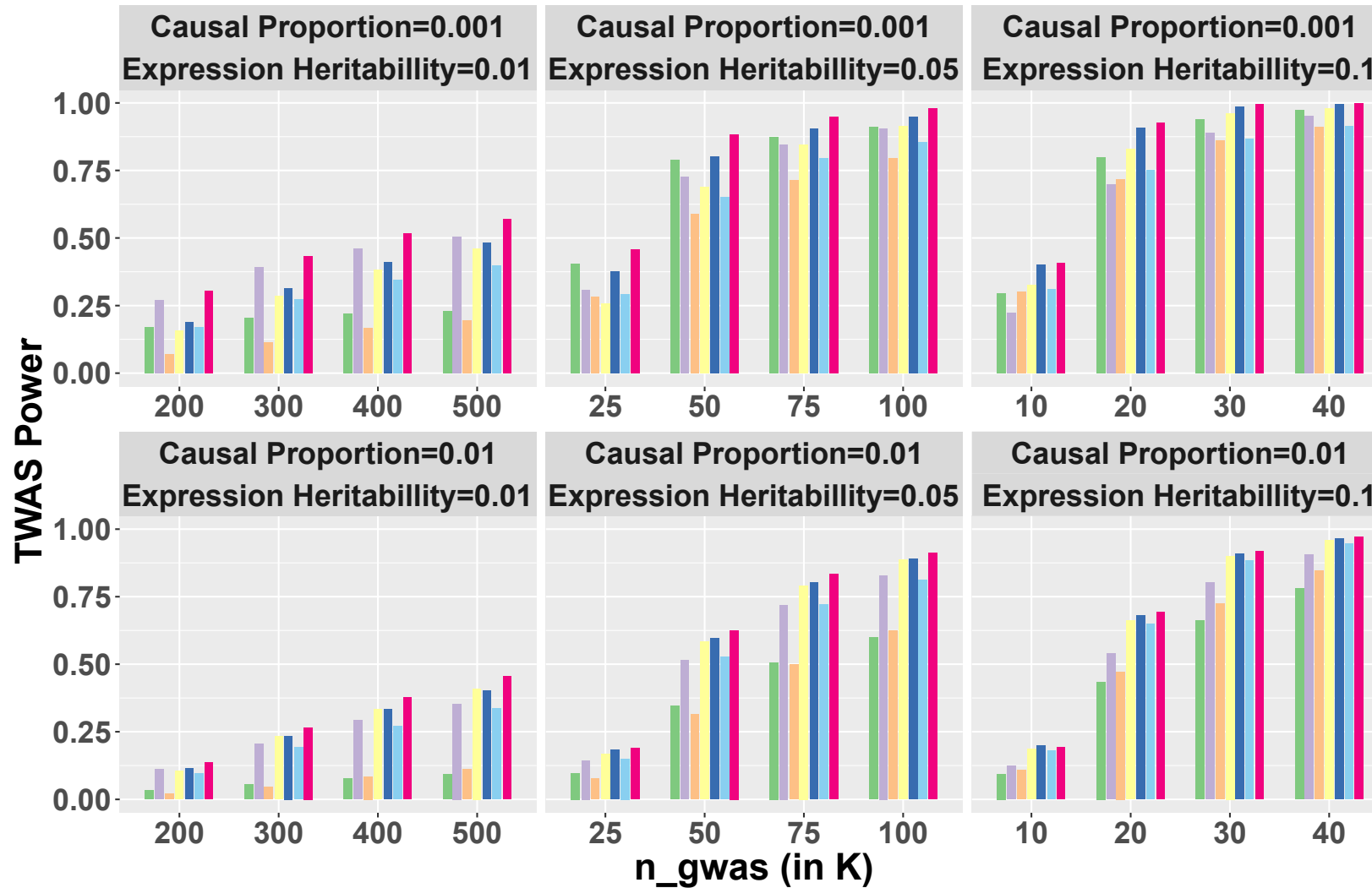
- 1894 WGS European subjects
 - Randomly select 30% for training and 70% for prediction to mimic real training sample size
- Genotypes: randomly select 500 genes and their cis-SNPs
 - Randomly select causal eQTLs: $p_{causal} = (0.001, 0.01)$
 - Proportion of expression variance explained by causal eQTLs: $h_e^2 = (0.01, 0.05, 0.1)$
- Generated GWAS Z-scores: $Z \sim MVN(R_g \mathbf{w} \sqrt{n_{gwas} * h_p^2}, R_g)$
 - $h_p^2 = 0.025$: Proportion of phenotypic variance explained by simulated expression ($X_g \mathbf{w}$)
 - n_{gwas} : GWAS sample size
 - R_g : Correlation matrix of standardized genotype matrix X_g

Simulation Results (Test R^2)

- 568 training samples
- 1326 test samples
- Imputation models fit using summary-level eQTL results from simulated training data
- **FUSION** uses individual-level reference eQTL data



TWAS Power with $h_p^2 = 0.025$, $\alpha = 2.5 \times 10^{-6}$



OTTERS has well-controlled Type I error

TWAS type I errors under 10^6 null simulations

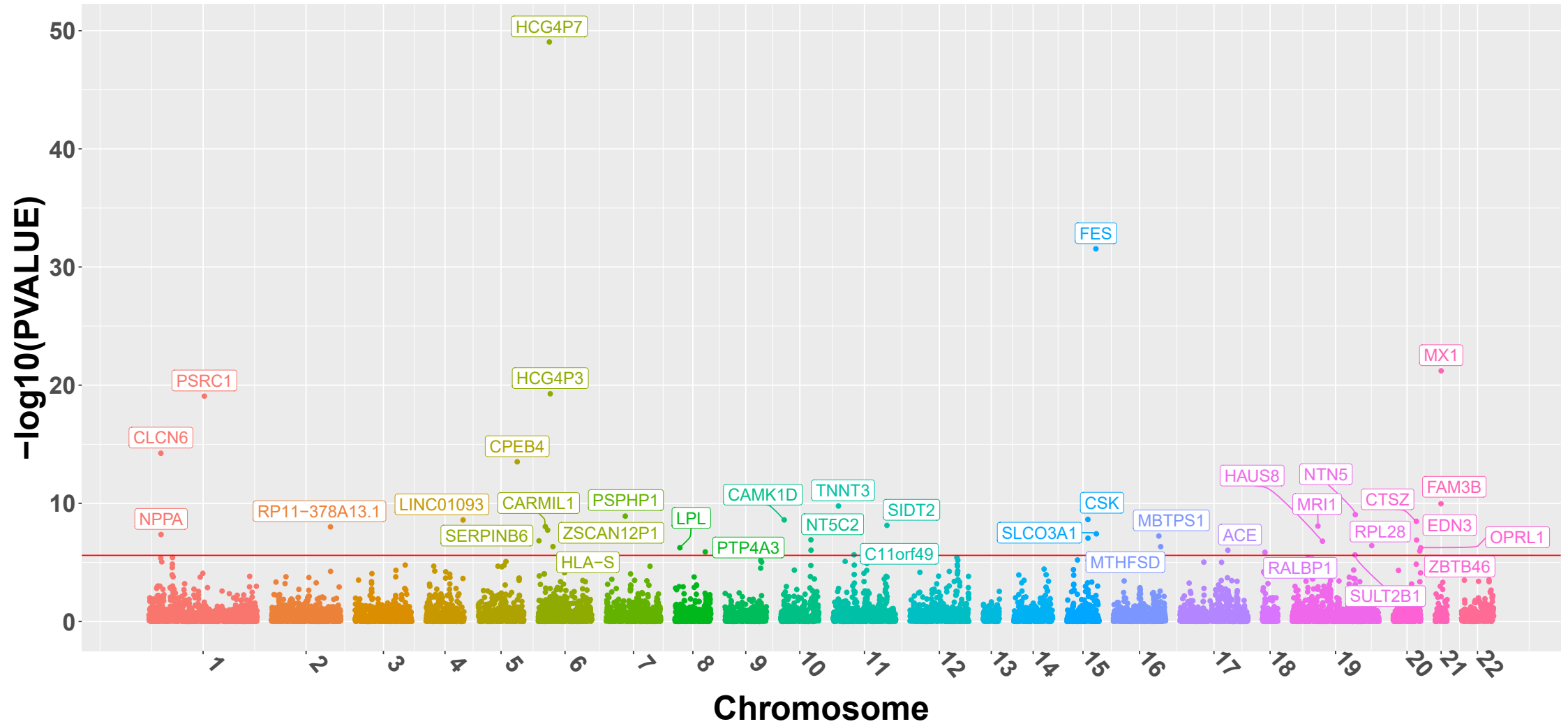
Significance Level	OTTERS Type-I Error
1.0×10^{-2}	1.00×10^{-2}
1.0×10^{-4}	9.30×10^{-5}
2.5×10^{-6}	2.00×10^{-6}

Type-I Error: how often OTTERS falsely detects an association when there is none

Apply OTTERS to study cardiovascular disease

- **Input data of OTTERS**
 - **eQTL summary-level statistics:** eQTLGen (n=31,684), 19K genes
 - **LD reference:** GTEx V8 WGS samples (n = 838)
 - **GWAS data:** UK Biobank summary-level GWAS results of cardiovascular disease (n=459,234; case fraction=0.319)
- **Results: association test between 19K genes and cardiovascular disease**
 - **Genes with OTTERS p-value < 0.05 after Bonferroni correction** were identified as genes significantly associated with cardiovascular disease

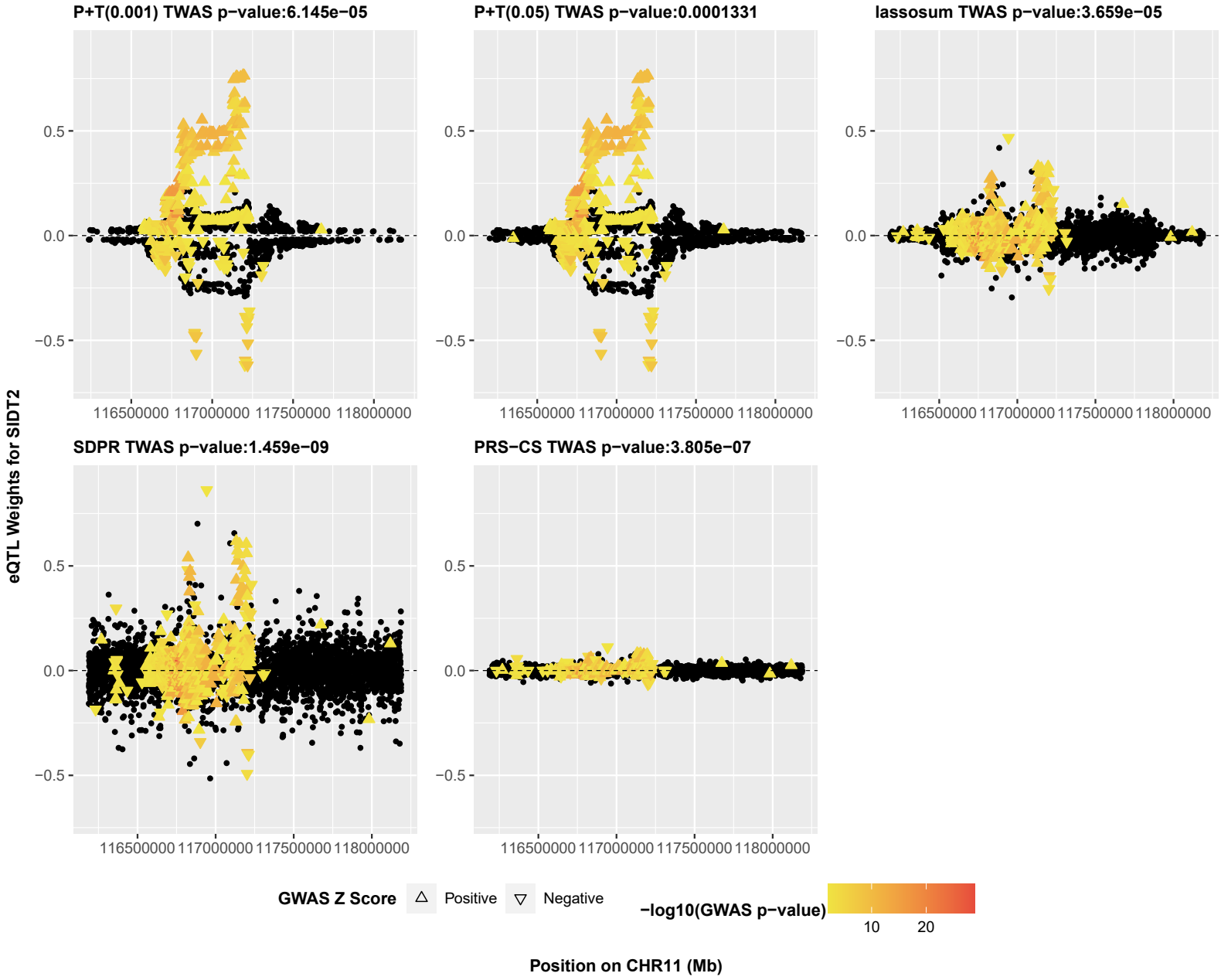
TWAS Results of Cardiovascular Disease by OTTERS



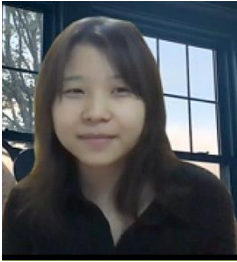
Apply OTTERS to study cardiovascular disease

- **OTTERS identified 38 genes associated with cardiovascular disease**
 - Only 8 were previously found by FUSION using GTEx V6 whole-blood samples (n=338).
 - 13 genes were not detected by FUSION in whole-blood samples,
 - but were identified by FUSION in other GTEx tissue types (such as artery, adipose visceral).
 - 17 genes were novel risk genes.
- **OTTERS leveraging large-scale summary-level eQTL data** can identify novel risk genes missed by **individual-level TWAS methods using reference panel with smaller sample sizes.**

Example Scatter Plots of eQTL Weights

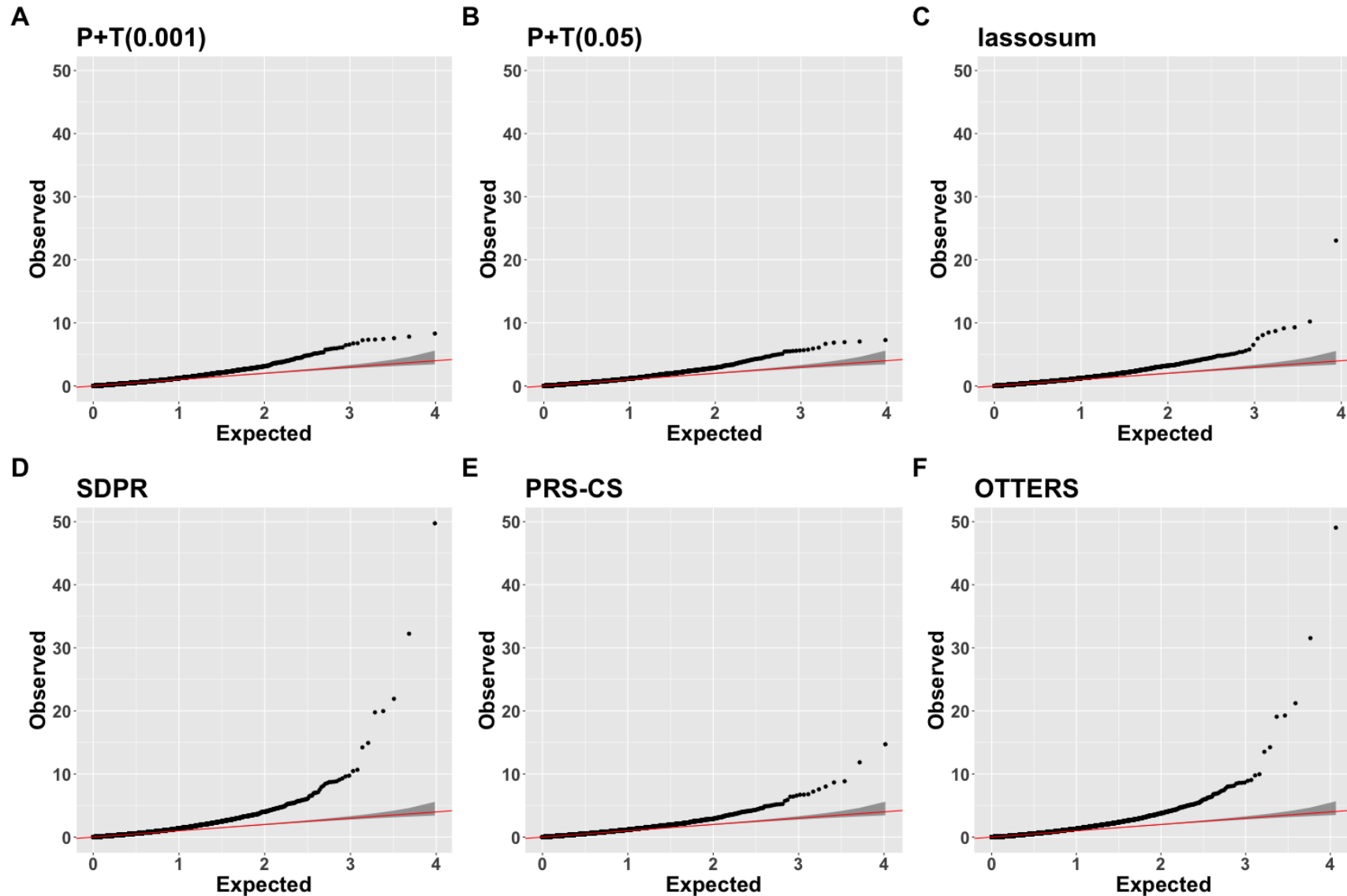


Dr. Michael
Epstein

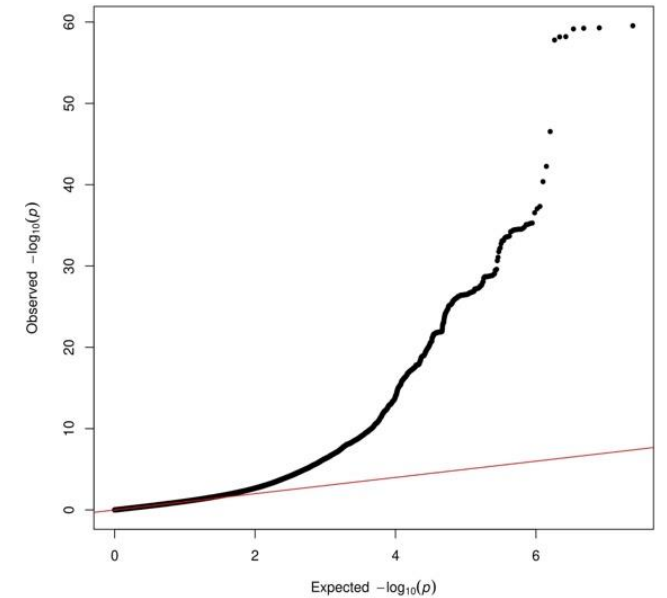


Qile Dai

TWAS Results of Cardiovascular Disease: Q-Q Plot



GWAS Results of Cardiovascular Disease in UKBB



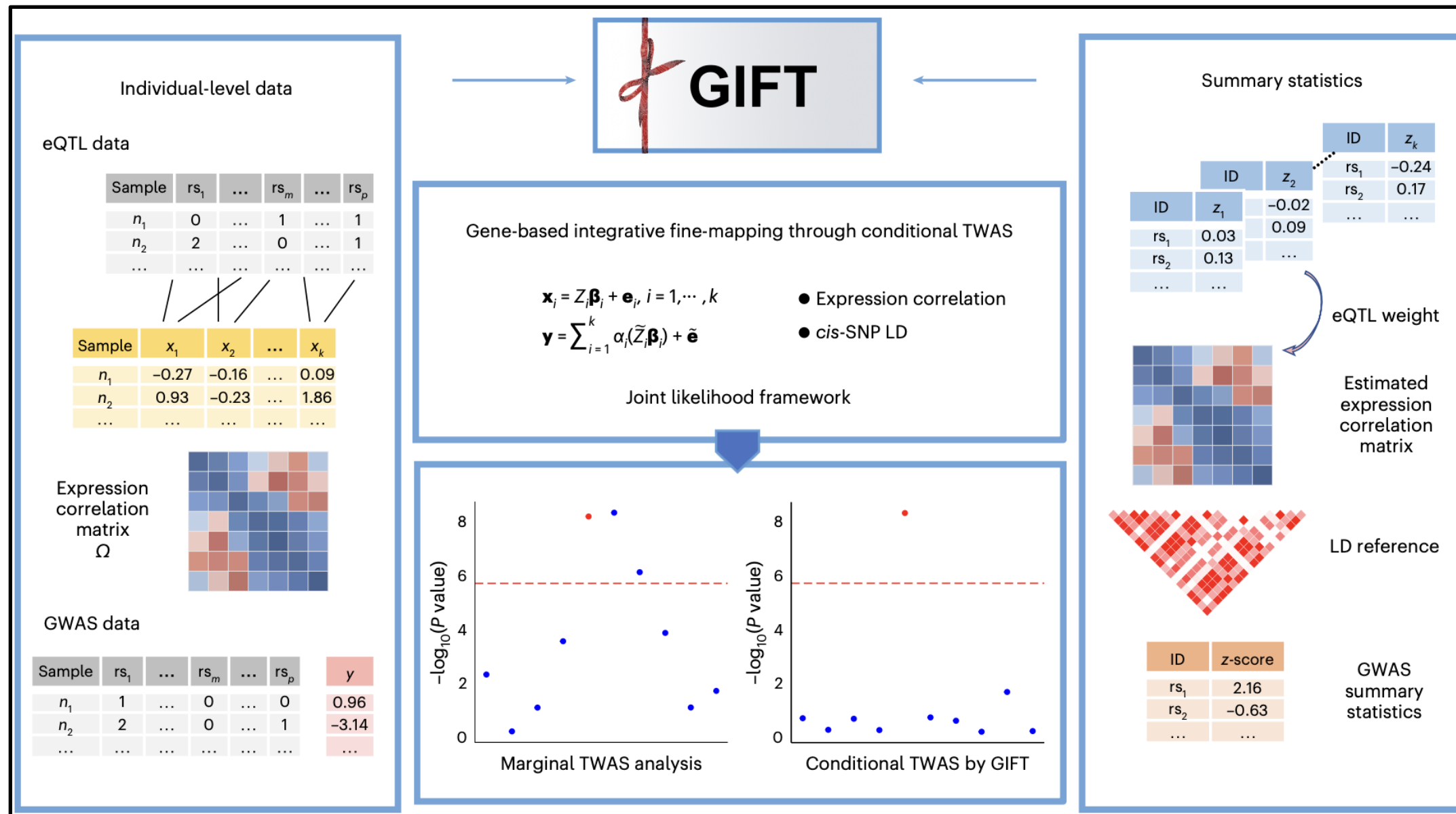
Fine-mapping TWAS

- TWAS tests genes-trait association independently for all transcriptome-wide genes
- TWAS results often contain a cluster of significant genes in the same region
 - LD causes inflation in GWAS summary data
 - Test SNPs are overlapped in nearby genes
 - Nearby GReX are correlated
- Fine map for potential “causal” or independent significant TWAS risk gene

Fine-mapping TWAS Methods

- **FOCUS** (fine-mapping of causal gene sets; Mancuso N. et al. Nat. Genet. 2019): Extends the Bayesian sparse model used in GWAS fine-mapping to TWAS
- **FOGS** (fine-mapping of gene sets; Wu C. et al. Hum. Genet. 2020): Performs conditional SNP association analysis and then aggregates the conditional SNP-trait associations into gene-trait associations
- **GIFT** (gene-based integrative fine-mapping through conditional TWAS; Liu L. et al. Nat. Genet. 2024): Models the GReX of all genes residing in the focal region and carries out TWAS conditional analysis in a maximum likelihood framework.

GIFT: Gene-based Integrative Fine-mapping through conditional TWAS



Statistical Model of GIFT with Individual-level eQTL and GWAS Data

- **Gene Expression Prediction Model:**

$$\mathbf{x}_i = \mathbf{Z}_i \boldsymbol{\beta}_i + \mathbf{e}_i, i = 1, \dots, k,$$

- $\mathbf{x}_i$: Expression level of the i^{th} gene
- $\mathbf{Z}_i$: Genotype matrix of cis-SNPs for gene i
- $\boldsymbol{\beta}_i$: Effect sizes of the cis-SNPs
- e_i : Residual errors.

- **Gene-Trait Association Model:** $\mathbf{y} = \sum_{i=1}^k \alpha_i (\tilde{\mathbf{Z}}_i \boldsymbol{\beta}_i) + \tilde{\mathbf{e}},$

- $\mathbf{y}$: Trait outcome
- α_i : Effect sizes of the GReX of gene i on the trait
- $\tilde{\mathbf{Z}}_i$: Genotype matrices in the test GWAS data
- $\tilde{\mathbf{e}}$: Residual error

Note that the notations here are matched with the GIFT paper, which is different from the notations in our previous slides.

With eQTL and GWAS Summary Data

- **Gene Expression Model:** $\hat{\mathbf{z}}_{x_i} = \sqrt{(n_1 - 1)}\Sigma_{1i}\boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_{x_i}, i = 1, \dots, k,$
 - $\widehat{\mathbf{Z}}_{x_i}$: eQTL Z-score vector for the i_{th} gene
 - Σ_{1i} : SNP-SNP correlation matrix for the i_{th} gene's with dimension $p_i \times p_i$
 - β_i : Marginal effect sizes of cis-SNPs on the i_{th} gene's expression.
 - ε_{x_i} : Residual errors for the i_{th} gene
- **Gene-Trait Model:** $\hat{\mathbf{z}}_y = \sqrt{(n_2 - 1)}\Sigma_2(\alpha_1\boldsymbol{\beta}_1^T, \dots, \alpha_k\boldsymbol{\beta}_k^T)^T + \boldsymbol{\varepsilon}_y,$
 - $\widehat{\mathbf{Z}}_y$: Stacked GWAS Z-score vector of all genes
 - Σ_2 : SNP-SNP correlation matrix for stacked cis-SNPs across all genes with dimension $\sum_{i=1}^k p_i \times \sum_{i=1}^k p_i$
 - $\alpha_i\beta_i$: Product of GReX effect size and SNP effect sizes for the i_{th} gene
 - ε_y : Residual errors in GWAS

GIFT

- Test $H_0: \alpha_i = 0, i = 1, \dots, k$
- Derive the likelihood framework through joint modeling the gene expression model and the gene-trait model
 - Accounts for the uncertainty of eQTL weight estimation
- Uses Expectation Maximization algorithm to maximize the joint likelihood to estimate $\{\alpha_i, i = 1, \dots, k\}$ and calculates the maximum likelihood under the full model
- Calculates the maximum likelihood under a series of reduced model, assuming the test $H_0: \alpha_i = 0$.
- Conducts likelihood ratio tests

Apply GIFT to Study Blood Pressure and Lipid Traits with the UK Biobank Data

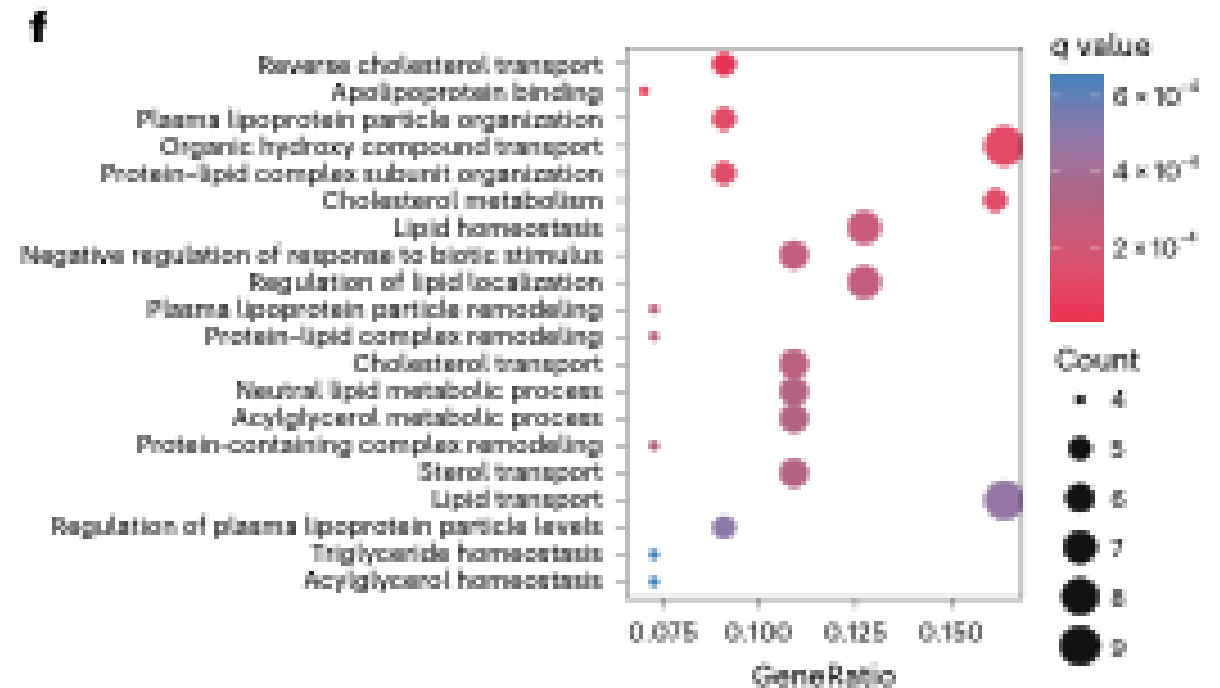
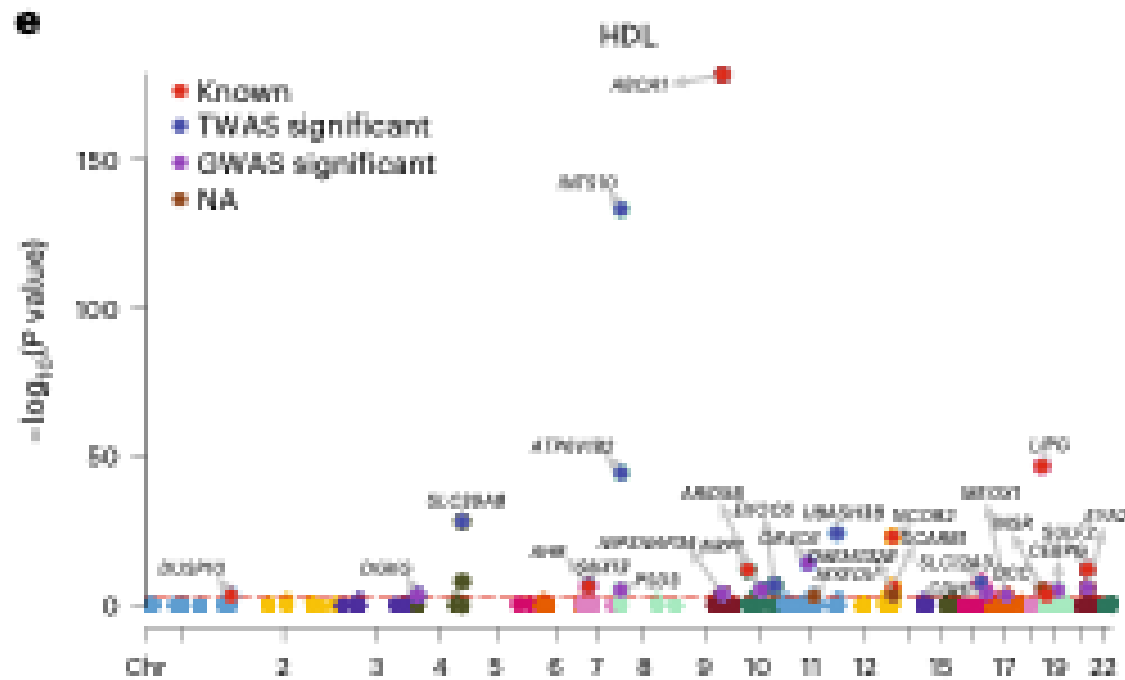
- Apply GIFT to studying 15,577 genes residing in 1,533 independent LD regions

Table 1 | Summary results of fine-mapping in the real data

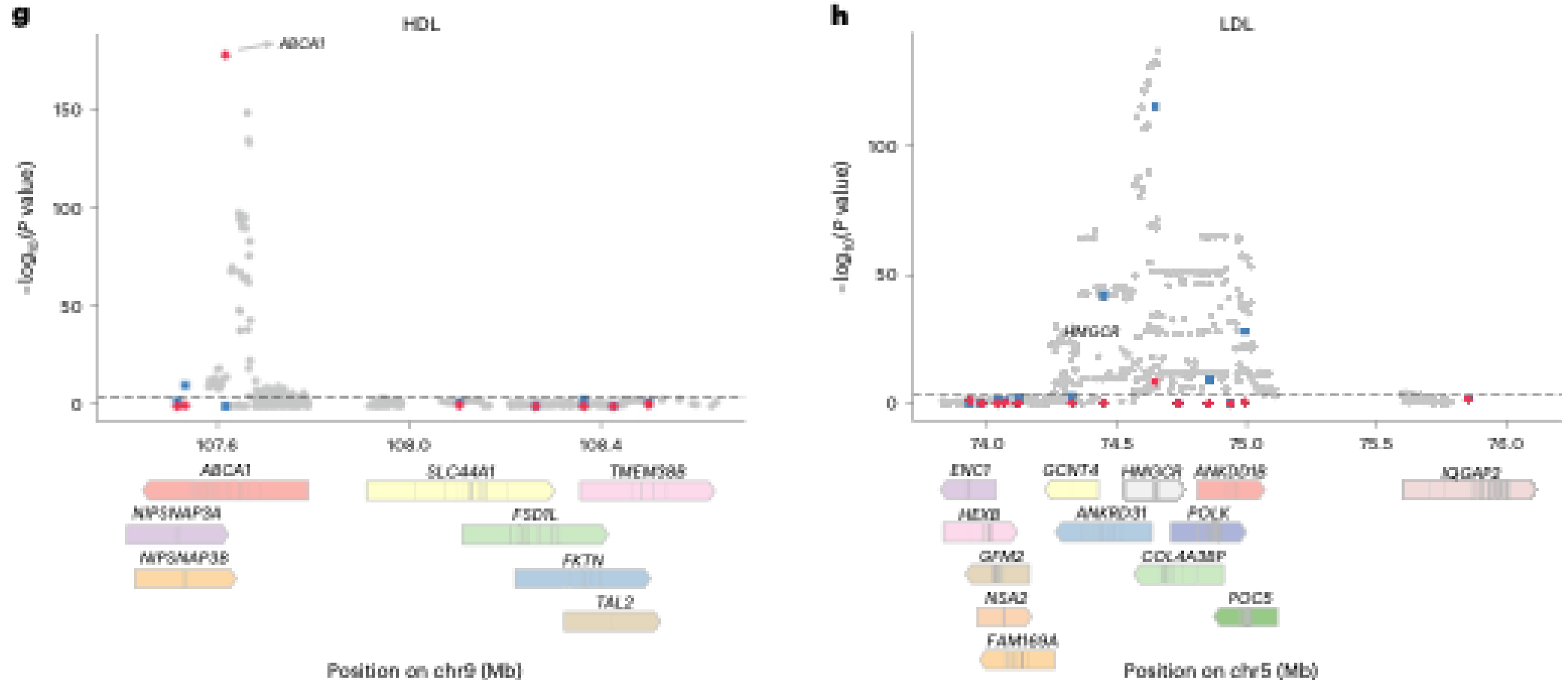
Trait	GWAS risk regions	Significant TWAS genes	TWAS risk regions	GWAS risk regions with TWAS significant genes	GIFT	FOCUS	FOGS	MY-IWAS
SBP	147	148	85	53	7	72	350	409
DBP	191	181	87	67	13	75	356	691
TC	147	286	108	82	19	136	599	804
HDL	231	473	152	120	28	178	937	1,320
LDL	122	226	93	72	16	116	455	679
TG	200	393	124	96	31	149	753	922

The table shows the number of discoveries for each of the six traits (rows) in TWAS fine-mapping analysis carried out by each of the four methods (columns). A GWAS risk region (first column) is defined as an LD block that harbors at least one genome-wide significant SNP ($P < 5 \times 10^{-8}$). A significant TWAS gene (second column) is defined as a gene with a marginal TWAS $P < 0.05/15,577$. A TWAS risk region (third column) is defined as an LD block that harbors at least one marginal TWAS significant gene. The last four columns list the number of genes discovered by each of the four methods for the six traits. We used an empirical FDR threshold of 0.05 to declare significance for all methods in the fine-mapping analysis.

GIFT Results for HDL



Example GIFT Results for HDL and LDL



Web Resources

- **OTTERS**
 - <https://github.com/daiqile96/OTTERS>
- **GIFT**
 - <https://yuanzhongshang.github.io/GIFT/>
- **Public Sources of Summary-level eQTL Data**
 - GTEx (Genotype-Tissue Expression Project): <https://gtexportal.org>
 - eQTLGen Consortium: <https://www.eqtlgen.org>
 - eQTL Catalogue: <https://www.ebi.ac.uk/eqtl/>
 - DepMap (for cancer cell lines): <https://depmap.org>